MIE353 Assignment 2 – Part A

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Question 1

Q1. There are 5 entity classes in the schema: `Course`, `Professor`, `Student`, `Teaches`, and `Transcript`. These classes are not written as separate Java files in the project. Manifold generates them automatically, one for each table in the `cms-ddl.sql` file. For example, `Main.java` uses the `Student` class when it calls `Student.fetch(9002)` to get a student by their ID.

Q2. The `cms-ddl.sql` file defines the structure of the database. It has the `CREATE TABLE` statements for the five tables, and it sets each table's columns, data types, primary keys, and foreign keys. It also adds some rules, like grades having to be between 0 and 100 and professor emails having to be unique. The file is used in `CMS.dbconfig` through the line `"dbDdl": "/config/cms-ddl.sql"`. Manifold uses it to set up the tables in the database and to generate the Java entity classes, so the SQL queries can be checked against the schema when the code compiles.

Q3. The transcript table has two foreign keys:

- `studentId`, which references `studentId` in the `student` table
- `courseId`, which references `courseId` in the `course` table

This means every transcript record has to belong to an existing student and an existing course. These two columns together also make up the primary key, so a student can only have one grade for each course.

Q4. The `FindStudent` query has one parameter, `:studentId`. The query is `SELECT * FROM student WHERE studentId = :studentId`, which returns the student with the given ID. In `Main.java`, it is called with one value: `FindStudent.fetchOne(9002)`.

Q5. The database is a DuckDB database that runs in memory. In `CMS.dbconfig`, the URL is just `jdbc:duckdb:` with no file path, and `inMemory` is set to true. Because of this, the database is not saved to a file on disk and only exists while the program is running. The actual data is kept in CSV files in the `src/main/resources/data` folder. Each time the program runs, the `Load` queries read these files into the database using the `COPY` command. Results can also be saved back out as CSV files, like the `SaveStudents` query does.